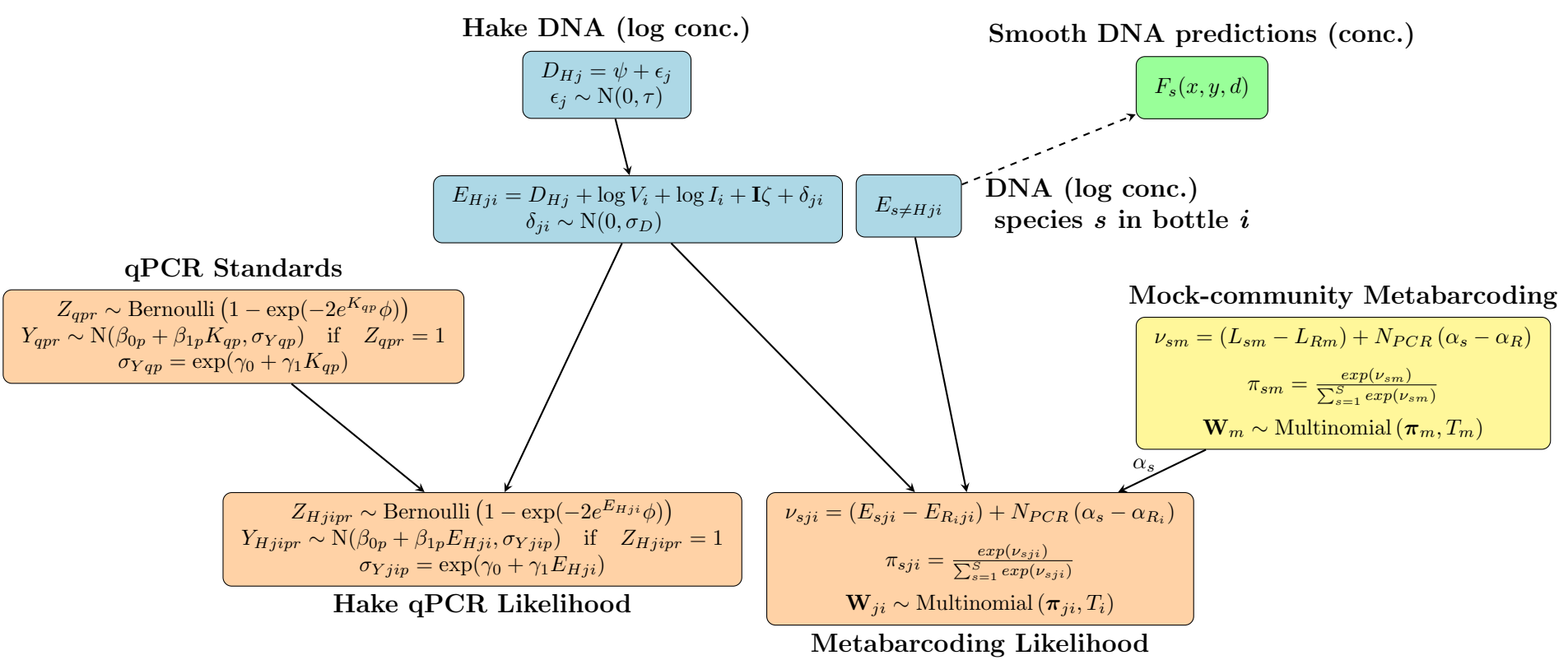




Index

s = species (H = Hake)
 j = location (northing, easting, depth)
 x = northing
 y = easting
 d = depth
 i = biological replicate
 r = technical replicate
 p = qPCR plate
 m = mock community sample
 q = qPCR standard aliquote sample
 R = metabarcoding reference species

Parameters and derived quantities

ψ = mean hake eDNA concentration (log(copies/ μ L))
 δ = biological replicate random effect
 V_i = volume filtered and aliquote dilution offset
 ϕ = qPCR amplification parameter
 σ_Y = standard deviation of qPCR Ct values
 β_{0p}, β_{1p} = intercept and slope defining mean qPCR - DNA concentration relationship
 γ_0, γ_1 = intercept and slope defining standard deviation of qPCR - DNA concentration relationship
 α_s = amplification efficiency relative to reference species
 π = predicted sequence proportions at the end of PCR
 ζ = ethanol wash effect
 τ = standard deviation among biological replicates

Observations

Z = qPCR amplification (yes=1; no=0)
 Y = qPCR Ct values
 K = known DNA conc. (log(copies/ μ L))
 L = known DNA conc. (log(copies/ μ L))
 W = metabarcoding sequencing reads
 T = total number sequencing reads
 V = volume filtered and aliquote dilution offset
 I = aliquote dilution factor

States

D = DNA conc. at a station-depth (log(copies/ μ L))
 E = DNA conc. in a biological replicate (log(copies/ μ L))
 F = Smoothed DNA conc. (copies/ L)